

Problem 1. You decided to examine how macroeconomic conditions affect drug overdose deaths. Suppose you ran the following simple regression using county-level data and obtained the results. The dependent variable, Overdose Death Rate, represents the number of deaths per 100,000 people, and the variable of interest, Unemployment Rate, is measured in percentage.

$$\text{Overdose Death Rate}_i = \beta_0 + \beta_1 \text{Unemployment Rate}_i + u_i$$

- (a) Interpret the coefficients. (Assume that $\beta_0, \beta_1 > 0$.)

β_0 : Expected odd rate when ue rate is zero.

β_1 : if ue rate increases by 1pp, the expected odd rate increases by β_1 .

- (b) Do you think the regression suffers from omitted variable bias? If so, give one example of a confounder and predict the direction of the omitted variable bias due to the confounder.

Poverty rate

True : odd rate = $\beta_0 + \beta_1 \text{UE rate} + \beta_2 \text{Poverty rate} + u_i$

Aux : Pov rate = $\delta_0 + \delta_1 \text{UE rate}$

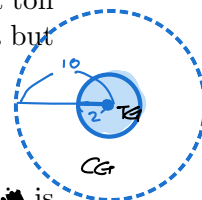
Bias = $\beta_2 \cdot \delta_1 > 0$ $\leftarrow$ explain $\Rightarrow$ An upward bias

- (c) Hollingsworth et al. (2017) use a panel data set and run the same regression model but with county and year fixed effects. According to their result, the estimated coefficient of the Unemployment Rate is (0.22) with the standard error of 0.05. Interpret the result.

if a county's UE rate increases by 1pp.

the odd rate rises by 0.22, controlling for the year FEs.

Problem 2. Currie and Walker (2011) examine the impact of introducing E-ZPass, an electronic toll collection system that reduced traffic congestion at toll plazas and, in turn, lowered pollution levels. They assess how this change affected infant health by comparing mothers living near toll plazas. In their DD design, the treatment group consists of mothers residing within 2 km of a toll plaza, while the control group includes those living between 2 km and 10 km from a toll plaza but still near a highway.



$$Y_{it} = \beta_0 + \beta_1 \text{E-ZPass}_{it} + \beta_2 \text{Close}_{it} + \beta_3 (\text{E-ZPass}_{it} \times \text{Close}_{it}) + u_{it}$$

E-ZPass is an indicator equal to one if the closest toll plaza has implemented E-ZPass; close is an indicator equal to one if the mother lived within 2 km of a toll plaza.

Fill in the table in terms of $\beta_0, \beta_1, \beta_2$, and β_3

EZ = ϕ

EZ = 1

	close = 1 <2 km E-ZPass	close = ϕ >2 km and <10 km E-ZPass
Low birth weight (before)	$E(y c=1, EZ=\phi) = \beta_0 + \beta_2$	$E(y c=\phi, EZ=\phi) = \beta_0$
Low birth weight (after)	$E(y c=1, EZ=1) = \beta_0 + \beta_1 + \beta_2 + \beta_3$	$E(y c=\phi, EZ=1) = \beta_0 + \beta_1$

The summary statistics show the following:

	<2 km E-ZPass	>2 km and <10 km E-ZPass
Low birth weight (before)	0.082	0.089
Low birth weight (after)	0.078	0.092
Difference, after-before	$0.078 - 0.082 = -0.004$	$0.092 - 0.089 = 0.003$

$\Rightarrow \beta_{DD}$

Source: Currie and Walker (2011) Table 1.

$= -0.004$
 $- 0.003$

(a) Find the causal effect of E-ZPass on low birth weight.

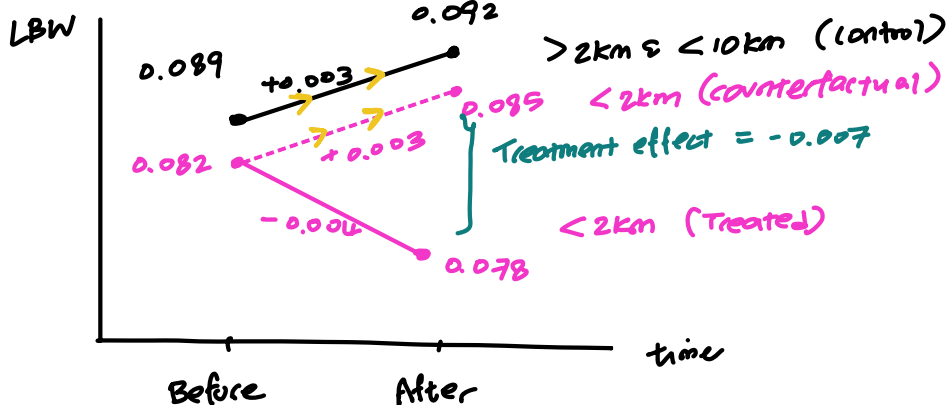
$$E(y|c=1, EZ=1) - E(y|c=1, EZ=\phi) = -0.004$$

$$E(y|c=\phi, EZ=1) - E(y|c=\phi, EZ=\phi) = 0.003$$

$$\therefore \text{Causal effect} = \Delta_{<2\text{km}} - \Delta_{>2\text{km} \& <10\text{km}} = -0.004 - 0.003 = -0.007$$

(b) In the space below, draw the following:

- Low birth weight before and after for mothers living within 2 km of an E-ZPass.
- Low birth weight before and after for mothers living between 2 km and 10 km of an E-ZPass.
- Low birth weight for mothers living within 2 km of an E-ZPass in the absence of E-ZPass adoption.
- The treatment effect.



Problem 3. In the U.S., individuals become eligible for Medicare when they turn 65. Many economists have used this age threshold to study the effect of health insurance on the utilization of health care services. Suppose we have monthly insurance claims data for individuals from 12 months before to 12 months after their 65th birthday.

(a) Fill out the following:

- Treatment: *being eligible for Medicare*
- Outcome: *use of healthcare services*
- Running variable: *Age*
- Cutoff: *65*
- Treated individuals: *Age ≥ 65*
- Control individuals: *Age < 65*

(b) Write out an equation to estimate an RD design with different slopes before and after the cutoff. Draw an RD graph assuming that all the coefficients are positive. Label the following:

- The treatment and control groups, the x- and y-axes, and the cutoff
- $E[Y|\text{Treat} = 1] = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)(\text{Age} - 65)$
- $E[Y|\text{Treat} = 0] = \beta_0 + \beta_2(\text{Age} - 65)$
- $E[Y|\text{Treat} = 1]$ at the cutoff $\beta_0 + \beta_1$
- $E[Y|\text{Treat} = 0]$ at the cutoff β_0
- The treatment effect β_1

$$y_i = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \overbrace{(\text{Age} - 65)}^{\text{distance}} + \beta_3 \overbrace{[\text{Treat} \times (\text{Age} - 65)]}^{\text{distance}} + u_i$$

